Grade 10 – Term 1: C8 Classroom Activity

Goal: Compare decision criteria and solve numerical conditions that change a recommendation. All profits and monetary parameters are in thousand US dollars; larger profits are better. Use the supplied summaries for Task 1. For Task 2, show your inequality, boundary ties, updated calculations, and numerical justification. Unless stated otherwise, probabilities and other payoffs stay unchanged. Compute regrets using eligible alternatives only. Do not assume missing probabilities.

Guided Problems

1. Cold-storage cooperative. A produce cooperative chooses a storage system for the harvest season.

Alternative Probability	Strong demand 0.4	Normal demand 0.4	Weak demand 0.2
A_1: New warehouse	96	48	-12
A_2: Modular storage	72	60	18
A_3: Rented storage	48	42	34

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	96	-12	46	55.2
A_2	72	18	24	56.4
A_3	48	34	48	42.8

Task 1. Use the supplied summary to identify all four original recommendations and their deciding quantities and values. Report all ties; mark EMV/Bayes unavailable if needed.

Criterion	Alternative(s)	Deciding quantity and value
Maximax		
Maximin		
Minimax Regret		
EMV/Bayes		

Task 2. The lender requires profit of at least 18 in every state. A spoilage-prevention grant adds an integer s, $0 \le s \le 24$, only to A_2's weak-demand profit. Find the smallest s making Maximin strictly prefer A_2 over A_3. Solve an inequality, report the boundary tie, and identify eligible plans throughout the range. At that grant, complete a new summary for eligible plans and give all four recommendations with deciding values. Recompute regret using only eligible plans.

Working: threshold inequality, eligibility, and boundary tie.

Updated summary (label each row with its alternative; leave unused rows blank).

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV

Updated recommendations, deciding values, and numerical justification of changes.

2. Community laundry equipment. A community laundry selects an equipment package for the coming year. Listed net profits already include purchase costs.

Alternative Probability	High use 0.4	Normal use 0.4	Low use 0.2
A_1: Robotic package	120	54	6
A_2: Efficient package	84	66	30
A_3: Basic package	54	48	42

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	120	6	36	70.8
A_2	84	30	36	66
A_3	54	42	66	49.2

Task 1. Use the supplied summary to identify all four original recommendations and their deciding quantities and values. Report all ties; mark EMV/Bayes unavailable if needed.

Criterion	Alternative(s)	Deciding quantity and value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

Task 2. An additional annual maintenance fee f reduces A_1's profit by f in every state. Here f is an integer, $0 \leq f \leq 6$. Financing requires EMV of at least 64 after the fee. Find the smallest f making Bayes strictly prefer A_2 over A_1, and state the exact boundary tie (even if noninteger). Identify eligible plans throughout the range. At that fee, update the eligible summary, recalculate regrets, and report all four recommendations and deciding values. Deduct the fee once.

Working: threshold inequality, eligibility, and boundary tie.

Updated summary (label each row with its alternative; leave unused rows blank).

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

Updated recommendations, deciding values, and numerical justification of changes.

Standard Problems

3. Mountain excursion business. An excursion company chooses its seasonal booking capacity. A tour association may guarantee a minimum profit for its largest package.

Alternative Probability	Busy season 0.3	Normal season 0.5	Quiet season 0.2
A_1: Large package	110	38	-18
A_2: Medium package	82	62	14
A_3: Small package	58	50	32
A_4: Fixed contract	32	32	32

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	110	-18	50	48.4
A_2	82	14	28	58.4
A_3	58	32	52	48.8
A_4	32	32	78	32

Task 1. Use the supplied summary to identify all four original recommendations and their deciding quantities and values. Report all ties; mark EMV/Bayes unavailable if needed.

Criterion	Alternative(s)	Deciding quantity and value
Maximax		
Maximin		
Minimax Regret		
EMV/Bayes		

Task 2. The guarantee replaces A_1's quiet-season payoff of -18 with an integer g, $-18 \leq g \leq 32$. All four plans remain eligible. Find the smallest g making A_1 the unique Minimax Regret choice. Derive the maximum-regret expressions, solve the inequality, and report the boundary tie. Calculate the increase in the quiet-season payoff. At that g, update the summary and give all four recommendations and deciding values, including ties.

Working: threshold inequality, eligibility, and boundary tie.

Updated summary (label each row with its alternative; leave unused rows blank).

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV

Updated recommendations, deciding values, and numerical justification of changes.

4. Coffee roaster contracts. A coffee roaster chooses a wholesale contract. The initial probability of strong orders is 0.30; otherwise orders are weak.

Alternative Probability	Strong orders 0.3	Weak orders 0.7
A_1: Export contract	92	8
A_2: Regional contract	56	38
A_3: Local contract	44	42

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	92	8	34	33.2
A_2	56	38	36	43.4
A_3	44	42	48	42.6

Task 1. Use the supplied summary to identify all four original recommendations and their deciding quantities and values. Report all ties; mark EMV/Bayes unavailable if needed.

Criterion	Alternative(s)	Deciding quantity and value
Maximax		
Maximin		
Minimax Regret		
EMV/Bayes		

Task 2. A new forecast assigns probability p to strong orders and $1 - p$ to weak orders, with $0.30 \leq p \leq 0.80$. All contracts remain eligible. Solve the pairwise EMV inequalities to find the exact probability above which Bayes uniquely recommends A_1. State the Bayes recommendation on each side and at the boundary; check A_3 too. At $p = 0.50$, update the summary and give all four recommendations and deciding values. Justify which recommendations change.

Working: threshold inequality, eligibility, and boundary tie.

Updated summary (label each row with its alternative; leave unused rows blank).

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV

Updated recommendations, deciding values, and numerical justification of changes.

5. Solar installation service. A solar installer chooses a service plan. No reliable probabilities for demand are available.

Alternative	High demand	Normal demand	Low demand
A_1: Dedicated crew	88	30	-16
A_2: Shared crew	64	46	12
A_3: Outsource work	40	36	28

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	88	-16	44	Unavailable
A_2	64	12	24	Unavailable
A_3	40	28	48	Unavailable

Task 1. Use the supplied summary to identify all four original recommendations and their deciding quantities and values. Report all ties; mark EMV/Bayes unavailable if needed.

Criterion	Alternative(s)	Deciding quantity and value
Maximax		
Maximin		
Minimax Regret		
EMV/Bayes		

Task 2. A supplier offers a fixed rebate d that adds d to A_2's profit in every state; all costs are already included. Here d is an integer, $0 \leq d \leq 20$, and all plans remain eligible. Find the smallest d making Maximin strictly prefer A_2 to A_3. Solve the inequality and report the boundary tie. At that rebate, update the summary and all feasible recommendations with deciding values. Compute regret from the new state bests. Mark EMV unavailable; do not assign equal probabilities.

Working: threshold inequality, eligibility, and boundary tie.

Updated summary (label each row with its alternative; leave unused rows blank).

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV

Updated recommendations, deciding values, and numerical justification of changes.

Solutions

1. Cold-storage cooperative

Task 1. Maximax: A_1 (row maximum 96); Maximin: A_3 (row minimum 34); Minimax Regret: A_2 (maximum regret 24); EMV/Bayes: A_2 (EMV 56.4).

Task 2. A_1 fails because $-12 < 18$; A_2, A_3 remain eligible for every allowed s. Since $18 + s \leq 42 < 60$, the relevant minima are $18 + s$ and 34. Thus $18 + s > 34 \iff s > 16$, so the smallest integer is $s = 17$ (\$17,000). At $s = 16$, Maximin ties at 34. At $s = 17$, eligible state bests are $(72, 60, 35)$; regret rows are $(0, 0, 0)$ and $(24, 18, 1)$. Maximax changes through eligibility and Maximin changes through the grant.

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_2	72	35	0	59.8
A_3	48	34	24	42.8

Maximax: A_2 (row maximum 72); Maximin: A_2 (row minimum 35); Minimax Regret: A_2 (maximum regret 0); EMV/Bayes: A_2 (EMV 59.8).

All recommendations at the threshold boundary. Maximax: A_2 (row maximum 72); Maximin: A_2, A_3 (row minimum 34); Minimax Regret: A_2 (maximum regret 0); EMV/Bayes: A_2 (EMV 59.6).

2. Community laundry equipment

Task 1. Maximax: A_1 (row maximum 120); Maximin: A_3 (row minimum 42); Minimax Regret: A_1, A_2 (maximum regret 36); EMV/Bayes: A_1 (EMV 70.8).

Task 2. The original EMVs are 70.8, 66, and 49.2. Since the probabilities sum to one, $EMV(A_1) = 70.8 - f$. Eligibility requires $f \leq 6.8$, so A_1, A_2 qualify throughout $0 \leq f \leq 6$; A_3 never qualifies. Bayes changes when $66 > 70.8 - f \iff f > 4.8$. The boundary is a tie at $f = 4.8$; the smallest allowed integer is $f = 5$ (\$5,000). State bests at 5 are $(115, 66, 30)$, giving regret rows $(0, 17, 29)$ and $(31, 0, 0)$. Bayes changes to A_2; Maximin changes because A_3 is excluded.

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	115	1	29	65.8
A_2	84	30	31	66

Maximax: A_1 (row maximum 115); Maximin: A_2 (row minimum 30); Minimax Regret: A_1 (maximum regret 29); EMV/Bayes: A_2 (EMV 66).

All recommendations at the threshold boundary. Maximax: A_1 (row maximum 115.2); Maximin: A_2 (row minimum 30); Minimax Regret: A_1 (maximum regret 28.8); EMV/Bayes: A_1, A_2 (EMV 66).

3. Mountain excursion business

Task 1. Maximax: A_1 (row maximum 110); Maximin: A_3, A_4 (row minimum 32); Minimax Regret: A_2 (maximum regret 28); EMV/Bayes: A_2 (EMV 58.4).

Task 2. State bests remain $(110, 62, 32)$ throughout the range. Maximum regrets are $R_1 = \max\{24, 32 - g\}$, $R_2 = 28$, $R_3 = 52$, and $R_4 = 78$. A unique choice of A_1 requires $\max\{24, 32 - g\} < 28 \iff g > 4$. At $g = 4$, A_1, A_2 tie with maximum regret 28. The smallest integer is $g = 5$; the increase is $5 - (-18) = 23$ (\$23,000). At 5, A_1 has regrets $(0, 24, 27)$ and EMV $33 + 19 + 1 = 53$. Minimax Regret changes to A_1; Maximin retains its two-way tie.

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	110	5	27	53
A_2	82	14	28	58.4
A_3	58	32	52	48.8
A_4	32	32	78	32

Maximax: A_1 (row maximum 110); Maximin: A_3, A_4 (row minimum 32); Minimax Regret: A_1 (maximum regret 27); EMV/Bayes: A_2 (EMV 58.4).

All recommendations at the threshold boundary. Maximax: A_1 (row maximum 110); Maximin: A_3, A_4 (row minimum 32); Minimax Regret: A_1, A_2 (maximum regret 28); EMV/Bayes: A_2 (EMV 58.4).

4. Coffee roaster contracts

Task 1. Maximax: A_1 (row maximum 92); Maximin: A_3 (row minimum 42); Minimax Regret: A_1 (maximum regret 34); EMV/Bayes: A_2 (EMV 43.4).

Task 2. $E_1 = 8 + 84p$, $E_2 = 38 + 18p$, $E_3 = 42 + 2p$. Hence $E_1 > E_2 \iff p > 5/11$ and $E_1 > E_3 \iff p > 17/41$. Also $E_2 > E_3 \iff p > 1/4$, true throughout the allowed range. Bayes selects A_2 for $0.30 \le p < 5/11$, ties A_1, A_2 at $p = 5/11$ (EMV 508/11), and selects A_1 for $5/11 < p \le 0.80$. At the boundary $E_3 = 472/11 < 508/11$. At $p = 0.50$, EMVs are 50, 47, 43. Only Bayes changes: payoffs, extrema, and regrets are unchanged.

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	92	8	34	50
A_2	56	38	36	47
A_3	44	42	48	43

Maximax: A_1 (row maximum 92); Maximin: A_3 (row minimum 42); Minimax Regret: A_1 (maximum regret 34); EMV/Bayes: A_1 (EMV 50).

All recommendations at the threshold boundary. Maximax: A_1 (row maximum 92); Maximin: A_3 (row minimum 42); Minimax Regret: A_1 (maximum regret 34); EMV/Bayes: A_1, A_2 (EMV $\frac{508}{11}$).

5. Solar installation service

Task 1. Maximax: A_1 (row maximum 88); Maximin: A_3 (row minimum 28); Minimax Regret: A_2 (maximum regret 24); EMV/Bayes: unavailable (no probabilities).

Task 2. The shared crew has minimum $12 + d$, so $12 + d > 28 \iff d > 16$. The smallest integer is $d = 17$ ($17,000); at $d = 16$, A_2, A_3 tie under Maximin at 28. All three plans stay eligible. At 17, state bests are $(88, 63, 29)$, and regret rows are $(0, 33, 45)$, $(7, 0, 0)$, $(48, 27, 1)$. Maximin changes to A_2. No distribution is given, so all EMVs and the Bayes recommendation remain unavailable; equal probabilities would add unsupported information.

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	88	-16	45	Unavailable
A_2	81	29	7	Unavailable
A_3	40	28	48	Unavailable

Maximax: A_1 (row maximum 88); Maximin: A_2 (row minimum 29); Minimax Regret: A_2 (maximum regret 7); EMV/Bayes: unavailable (no probabilities).

All recommendations at the threshold boundary. Maximax: A_1 (row maximum 88); Maximin: A_2, A_3 (row minimum 28); Minimax Regret: A_2 (maximum regret 8); EMV/Bayes: unavailable (no probabilities).